Semester	T.E. Semester VI – Computer Engineering
Subject	E- waste
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E-Waste Management – Solution

1. Understanding the Problem

E-waste is considered the fastest-growing waste stream in the world mainly because of the rapid growth in technology consumption. People frequently purchase new smartphones, computers, televisions, and other electronic devices and replace older devices within a short period. Rapid technological development, shorter product lifespans, increasing consumer demand, and frequent upgrades are some of the major reasons for the rapid increase in e-waste.

E-waste is a significant problem for the environment and human health because electronic devices contain hazardous substances such as lead, mercury, cadmium, and other toxic chemicals. Improper disposal, dumping, or burning of e-waste can release these harmful substances into the environment, contaminating soil, water, and air. Exposure to these substances can cause serious health problems, affecting the nervous system, respiratory system, kidneys, and other organs.

At the same time, electronic waste contains valuable materials such as copper, aluminium, gold, and other metals. When e-waste is not properly collected and recycled, these valuable resources are wasted and the demand for new raw materials increases. Therefore, proper e-waste management is important for protecting the environment, human health, and natural resources.

2. The Solution Approach: AHP and ISM

AHP – Analytic Hierarchy Process

The main purpose of using the **AHP (Analytic Hierarchy Process)** method was to **identify and prioritize the challenges associated with e-waste management**. AHP compares different factors and assigns priorities or weights to them. This helps researchers determine which challenges are more important and require greater attention.

ISM – Interpretive Structural Modeling

The main purpose of using the **ISM (Interpretive Structural Modeling)** method was to **identify the relationships among the different challenges and arrange them in a hierarchical structure**. ISM helps determine which factors are the root or driving factors and which factors are dependent on other factors.

Difference Between AHP and ISM

AHP and ISM serve different but complementary purposes. **AHP is mainly used for ranking and prioritizing the challenges**, while **ISM is used for understanding the relationships and hierarchy among those challenges**.

In simple terms:

AHP → Which challenge is more important?

ISM → How are the challenges related to each other?

Thus, AHP helps identify the most important challenges, whereas ISM provides a structured model showing how different challenges influence one another.